AMATH 482: HOMEWORK 4

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Abstract. In this report, we describe the methods used to compare two models for classification of handwritten digits in the MNIST dataset. In order to find the most accurate model, we compared ridge classification and logistic regression across various subsets as well as on the full dataset and compared their performance with their classification accuracy scores.

1. Introduction and Overview

In this project, we test classification models in an attempt to accurately classify handwritten digits from the MNIST data set from Yann Lecun . The MNIST data set is a dataset consisting of handwritten digits that is commonly used to test machine learning algorithms. We have both a test training set consisting of digit images and labels, and a training set consisting of images and labels. We will use the training set to fit our model, then use the test set and labels to check the accuracy of our models. For this project, we will use both ridge and logical regression models and compare their accuracies when applied to the test set.

2. Theoretical Background

2.1. **Classification Models.** Classification is a supervised learning method related to regression in that it attempts to predict an output given an input [3]. The difference is that regression deals with continuous outputs, while classification deals with discrete outputs. In order to label discrete outputs, we use one-hot encoding in which each label is an independent vector [4]. This way, we avoid encoding false relationships into our labels. In this case, the output is the digit represented by the image, which can only be an integer between 0 and 9 and is therefore discrete. In this project, we used two classification models, ridge and logistic, that are described more fully below.

2.1.1. *Ridge Regression.* We stack all of out outputs, as one-hot encodings, into a matrix Y which allows us to derive the regression expression $min_W||ZW - Y||_F^2$ that is equivalent to solving a separate regression problem for each label [4].

Adding regularization, we get

$$min_W|||ZW - Y|_F^2 + \lambda||W||_F^2 \tag{1}$$

which allows us to only tune one hyperparameter λ and is a stronger constraint to avoid overfitting [4].

In ridge regression, we saw the ridge regression theorem:

Theorem 2.1 (Ridge Regression)**.** *If c satisfies $(Z^TZ + \lambda I)c = Z^Ty$ then c is minimizer of ridge regression $Z^TZ + \lambda I$ is invertible when $\lambda > 0$*

Date: March 10, 2026.

[1] This allowed us to always find a unique solution. In ridge classification, we rely on the same principle as our equation 1 will satisfy this theorem.

In order to get a classification from our ridge regression output, we compute the score vector $f(z) = z^T W$ and predict the label with the encoding closest to f(z), $\hat{y} = argmin||f(z) - e_j||^2$ which is equal to the label with the largest score [4].

2.1.2. *Logistic Regression.* The second model that we tested is logistic regression. This model relies on probabilities, which are continuous quantities, to determine the label most likely to correspond to the input. Since we want to use probabilities, we need a method of restricting the output variables to stay between 0 and 1, otherwise they will not be valid probabilities. To do this, we use a sigmoid function which is defined by $\sigma(x) = \frac{1}{1+e^{-x}}$. This makes our probability function $Pr(Y = y|x) = q(x) = \sigma(c_0 + \Sigma_{j=1}^{p} c_j x_j)$ [2]

In order to find the final label, we use Maximum Likelihood estimation, which selects the coefficients that maximize $\prod_{i \in I} q(x_i, c) \prod_{i \notin I}(1 - q(x_i, c))$ where $I = \{i : y_i = 1\}$ [2]

2.1.3. *Classification Accuracy.* In regression with continuous outputs, we commonly use mean squared error, or MSE, to evaluate a model's performance. However, this method of evaluation is not appropriate in the case of classification as the outputs are not numerical and therefore we cannot apply numerical operations. Instead, we use the accuracy equation $\frac{1}{m}\Sigma_{i=1}^{m} 1_{(f(x_i)=y_i)}$ with the classification error represented by 1 - accuracy. This calculation utilizes an indicator function which is non-differentiable. Therefore, we cannot directly optimize the accuracy. As a result, when fitting parameters we optimize a continuous quantity, but when selecting hyperparameters, we generally prefer classification accuracy. We can use cross-validation to compare classification accuracies and attempt to optimize them [4].

3. Algorithm Implementation and Development

Our first step is to read the files containing the MNIST dataset and convert the file into arrays of images and labels. To do this, we use the method for parsing idx3-ubyte files provided in the homework helper file. Then, we are able to apply PCA and find the first 16 PCA modes, which we then plot.

3.1. **Determine PCA.** Next, we need to determine the number of PCA modes necessary to approximate 85 percent of the Frobenius norm. This value will be used in future steps.

3.2. **Function to Choose Subsets.** Our next step is to create functions that create subsets of the data consisting only of the given digits. These subsets include the labels and images in both the training and testing data, but only in the digits that we select. To create this method, we traverse the digit labels and keep any that are equal to one of the desired digits as well as the corresponding image in both the training and testing data.

3.3. **Ridge Classification Application.** The first classification model we want to test is ridge classification. To do this, we make a subset using the function defined in 3.2 consisting of 1 and 8. Then we standardize our data using StandardScaler from the sklearn library [5]. Then, we apply pca using the PCA modes that capture 85 percent of the norm as found in 3.1. To determine the value of alpha to be used in ridge classification, we perform cross-validation using RidgeClassifierCV and RepeatedKFold from sklearn [5]. Finally, we fit the fit the ridge classifier with optimized alpha to the train data, then predict from the test data and calculate accuracy with the accuracy score function from sklearn.metrics [5].

3.4. **Logistic Regression.** Next, we want to compare our accuracy using the ridge classifier to logistic regression. To start, we use our scaled x data after PCA has been applied from the previous step to fit a logistic regression model using the LogisticRegression function from sklearn with no regularization term [5]. Then, predict from the test data and calculate the accuracy score, once again using the accuracy score function.

3.5. **Testing with New Subsets.** To evaluate each of the model's accuracies more fully, we test them on different subsets and once again compare their accuracies. The subsets we tested were (3, 8), (2, 7), and all digits. We used the same methods discussed above in 3.3 and 3.4 on these subsets.

4. Computational Results

4.1. **PCA Modes.** After finding the first 16 PCA modes and showing them as images, we get the following image:

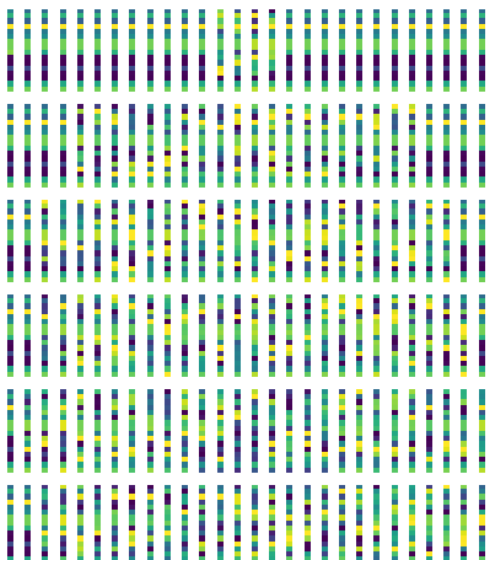

Figure 1. Image displaying the start of the image grid with the first 16 PCA modes for all digits.

After viewing this, we determined how many PCA modes would be necessary to capture 85 percent of the Frobenius norm and found that 21 modes were sufficient. Therefore, we will fit PCA with 21 modes to the data before our modeling steps to reduce dimensionality.

4.2. **(1, 8) Subset Classification Results.** For the first subset, consisting of digits 1 and 8, we found an accuracy of approximately 0.96707 with the ridge classifier, meaning we successfully classified 96.707 percent of the test set.

With logistic regression, we found an accuracy of 0.97568, or 97.568 percent. This is slightly higher meaning that in this subset, logistic regression slightly outperformed ridge classification.

4.3. **Other Subset Classification Results.** With the subset consisting of 3 and 8, we found an accuracy of 0.95187 with ridge classification and 0.94224 with logistic regression. Both of these values are lower than their respective performances in the previous subset, and in this subset ridge classification outperformed logistic regression, the opposite of the (1, 8) subset.

The final subset we tested was digits 2 and 7. In this subset we got an accuracy of 0.971 with ridge classification and an accuracy of 0.974 with logistic regression. As with the first subset,

logistic regression outperformed ridge classification, although the accuracies were extremely close. Both models also had a higher accuracy in this subset than in the (3, 8) subset.

4.4. **All Digits Classification Results.** Finally, we tested all digits and found an accuracy of 0.85172 for the ridge classifier and an accuracy of 0.87860 with logistic regression. These are the lowest accuracies out of any set tested for both models. Although they both performed worse than in previous subsets, logistic regression performed better than ridge regression.

To more clearly see the differences between models and subsets, we can compile our results into a table.

subset	ridge classification accuracy	logistic regression accuracy
(1, 8)	0.96707	0.97568
(3, 8)	0.95187	0.94224
(2, 7)	0.971	0.974
All digits	0.85172	0.87860

Table 1. Compiled accuracy results from all subsets and both models

5. Summary and Conclusions

In summary, we tested two classification models, ridge classification and logistic regression, to determine which model has the best performance in classifying handwritten digits from the MNIST dataset. We determined that in most subsets tested, including all digits, logistic regression outperformed ridge classification in terms of classification accuracy when applied to the test dataset that the model had not previously seen. Therefore, we can conclude that we should use logistic regression for related classification tasks, especially when dimensionality is limited as it was in this case.

6. Acknowledgements

The author would like to thank Professor Frank for providing the code for parsing idx3-ubyte files.

References

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